Practice Numerical Problems on Interference

1. A soap film of R.I. 1.32 and thickness 0.12 μm is illuminated by light incident at 30°. The reflected light shows a bright band corresponding to a wavelength of 586.3nm. Calculate order of interference of the band.
2. A parallel beam of light is incident at an angle of 30° on a film of refractive index 1.45. Find the colours from the visible spectrum (4000–7000 Å) which will be intensified in transmitted light? Thickness of the film is 5×10^{-5} cm.
3. A soap film is illuminated by monochromatic light of wavelength 6000 Å incident at certain angle. The film appears bright in reflected light. If its thickness is 1.5×10^{-5} cm and refractive index is 1.33, find the angle of incidence for minimum thickness.
4. A soap film of R.I. 1.31 and thickness 450nm is illuminated normally by light which shows a second dark band in reflected system. Calculate wavelength corresponding to this dark band.
5. White light falls normally on a soap film ($\mu = 1.33$) of thickness 3800 Å. Which wavelengths within the visible spectrum will be absent in the reflected light?
6. A parallel beam of light falls normally on an oil film of R.I. 1.2 having uniform thickness which is spread on water (R.I. 1.33). Brightness is obtained for wavelengths 5000 Å and 7500 Å and for no wavelength in-between. Find the thickness of the oil film.